

Experiment-4

Advanced Data Visualization using Matplotlib and Seaborn

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In [ ]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Step 1: Load dataset
df = pd.read_csv("netflix_titles.csv")
print("Dataset loaded successfully!")
print("\nFirst 5 Records:")
print(df.head())

# Step 2: Import Libraries and check dataset
print("\nDataset Shape:")
print(df.shape)
print("\nDataset Information:")
df.info()
print("\nMissing Values:")
print(df.isnull().sum())

# Step 3: Bar Chart - Movies and TV Shows
plt.figure(figsize=(8, 5))
sns.countplot(
    data=df,
    x="type"
)
plt.title("Movies vs TV Shows")
plt.xlabel("Type")
plt.ylabel("Count")
plt.tight_layout()
plt.show()

# Step 4: Line Chart - Content Added Over Years
year_data = df["release_year"].value_counts().sort_index()
plt.figure(figsize=(10, 5))
plt.plot(
    year_data.index,
    year_data.values,
    marker="o"
)
plt.title("Content Released by Year")
plt.xlabel("Release Year")
plt.ylabel("Number of Titles")
plt.grid(True)
plt.tight_layout()
plt.show()

# Step 5: Histogram - Release Year Distribution
plt.figure(figsize=(8, 5))
plt.hist(
    df["release_year"].dropna(),
    bins=20
)
plt.title("Distribution of Release Years")
plt.xlabel("Release Year")
plt.ylabel("Frequency")
```

```
plt.tight_layout()
plt.show()

# Step 6: Box Plot - Release Year by Type
plt.figure(figsize=(8, 5))
sns.boxplot(
    data=df,
    x="type",
    y="release_year"
)
plt.title("Release Year Distribution by Type")
plt.xlabel("Type")
plt.ylabel("Release Year")
plt.tight_layout()
plt.show()

# Step 7: Scatter Plot - Release Year vs Duration
# Extract numerical minutes from duration
df["duration_numeric"] = pd.to_numeric(
    df["duration"].str.extract(r"(\d+)")[0],
    errors="coerce"
)
scatter_data = df.dropna(
    subset=["release_year", "duration_numeric"]
)
plt.figure(figsize=(9, 5))
sns.scatterplot(
    data=scatter_data,
    x="release_year",
    y="duration_numeric",
    hue="type"
)
plt.title("Release Year vs Duration")
plt.xlabel("Release Year")
plt.ylabel("Duration")
plt.tight_layout()
plt.show()

# Step 8: Correlation Heatmap
numeric_data = df.select_dtypes(
    include="number"
)
correlation = numeric_data.corr()
plt.figure(figsize=(8, 6))
sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)
plt.title("Correlation Heatmap")
plt.tight_layout()
plt.show()

# Step 9: Customized Visualization
type_counts = df["type"].value_counts()
plt.figure(figsize=(8, 5))
bars = plt.bar(
    type_counts.index,
    type_counts.values
```

```
)
plt.title("Netflix Content Distribution")
plt.xlabel("Content Type")
plt.ylabel("Number of Titles")
for bar in bars:
    height = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width() / 2,
        height,
        str(int(height)),
        ha="center",
        va="bottom"
    )
plt.tight_layout()
plt.show()

# Step 10: Interpretation and Summary
print("\nVisualization Summary")
print("-" * 40)
print("1. The bar chart compares Movies and TV Shows.")
print("2. The line chart shows content distribution across release years.")
print("3. The histogram shows the distribution of release years.")
print("4. The box plot compares release-year distributions.")
print("5. The scatter plot shows the relationship between")
print("    release year and duration.")
print("6. The heatmap shows correlations between numerical variables.")
print("7. Customized charts improve readability and interpretation.")
print("\nExperiment completed successfully!")
```

Dataset loaded successfully!

First 5 Records:

	show_id	type	title	director	\
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	
1	s2	TV Show	Blood & Water	NaN	
2	s3	TV Show	Ganglands	Julien Leclercq	
3	s4	TV Show	Jailbirds New Orleans	NaN	
4	s5	TV Show	Kota Factory	NaN	

	cast	country	\
0	NaN	United States	
1	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	
2	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	
3	NaN	NaN	
4	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	

	date_added	release_year	rating	duration	\
0	September 25, 2021	2020	PG-13	90 min	
1	September 24, 2021	2021	TV-MA	2 Seasons	
2	September 24, 2021	2021	TV-MA	1 Season	
3	September 24, 2021	2021	TV-MA	1 Season	
4	September 24, 2021	2021	TV-MA	2 Seasons	

	listed_in	\
0	Documentaries	
1	International TV Shows, TV Dramas, TV Mysteries	
2	Crime TV Shows, International TV Shows, TV Act...	
3	Docuseries, Reality TV	
4	International TV Shows, Romantic TV Shows, TV ...	

	description
0	As her father nears the end of his life, filmm...
1	After crossing paths at a party, a Cape Town t...
2	To protect his family from a powerful drug lor...
3	Feuds, flirtations and toilet talk go down amo...
4	In a city of coaching centers known to train I...

Dataset Shape:
(8807, 12)

Dataset Information:

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 8807 entries, 0 to 8806

Data columns (total 12 columns):

#	Column	Non-Null Count	Dtype
0	show_id	8807 non-null	object
1	type	8807 non-null	object
2	title	8807 non-null	object
3	director	6173 non-null	object
4	cast	7982 non-null	object
5	country	7976 non-null	object
6	date_added	8797 non-null	object
7	release_year	8807 non-null	int64
8	rating	8803 non-null	object
9	duration	8804 non-null	object
10	listed_in	8807 non-null	object
11	description	8807 non-null	object

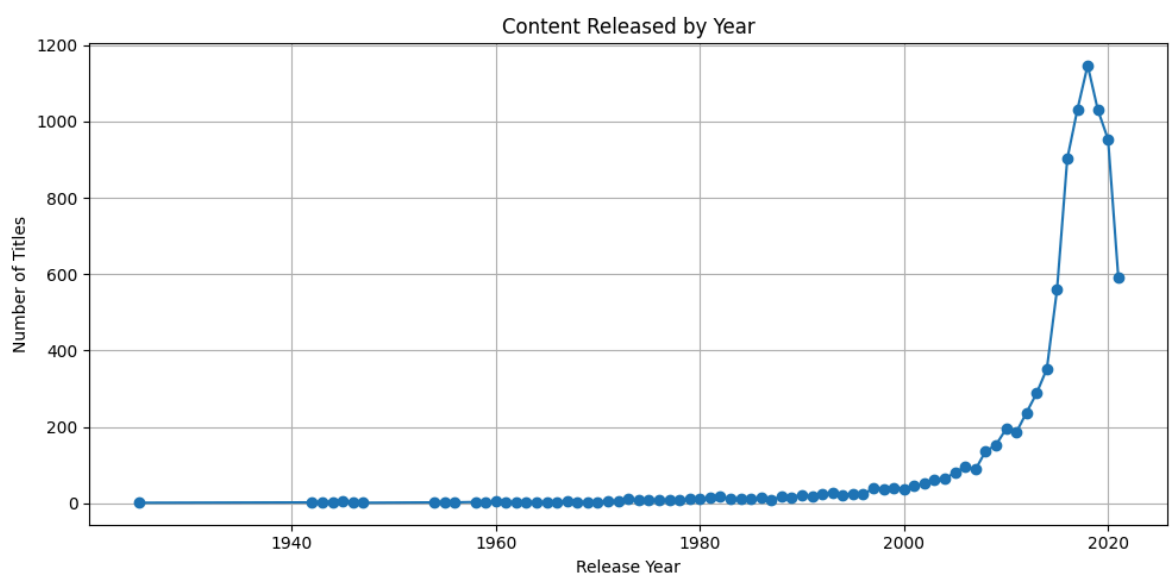
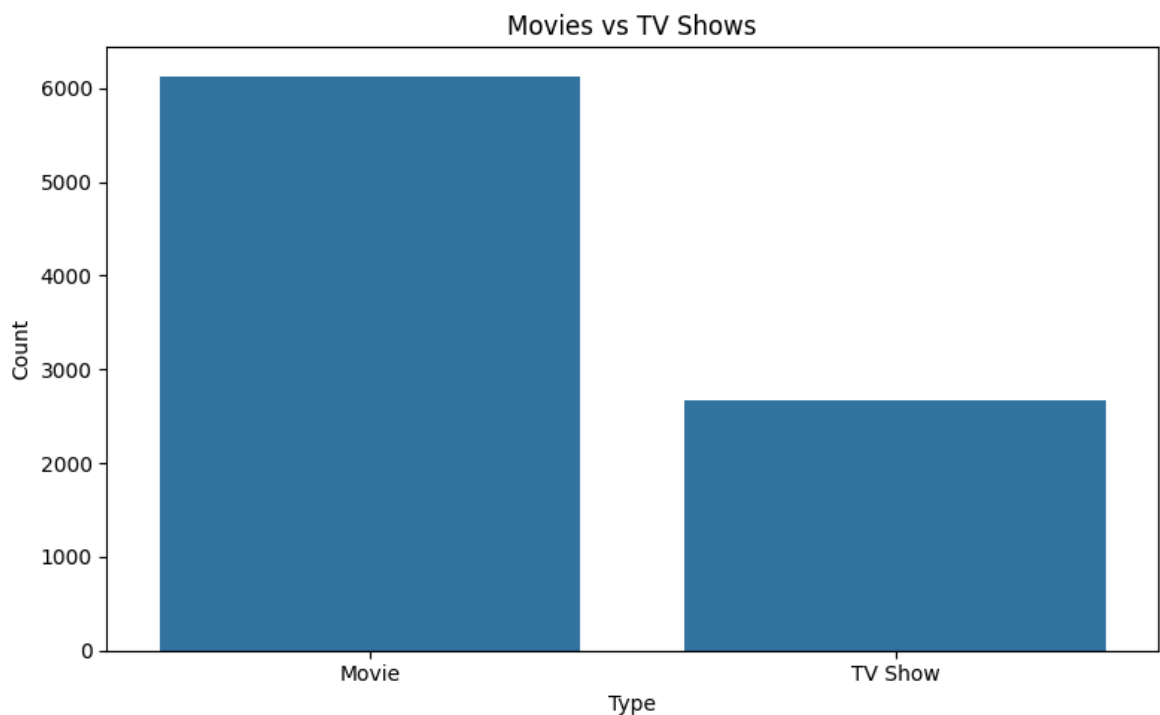
dtypes: int64(1), object(11)

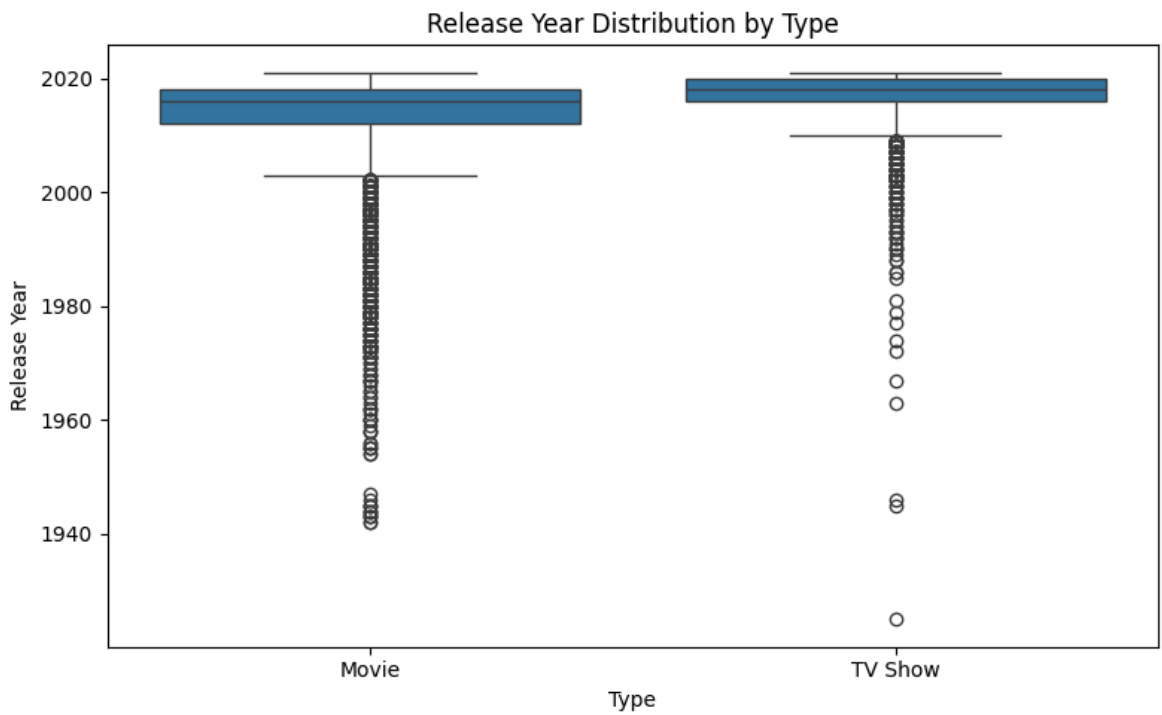
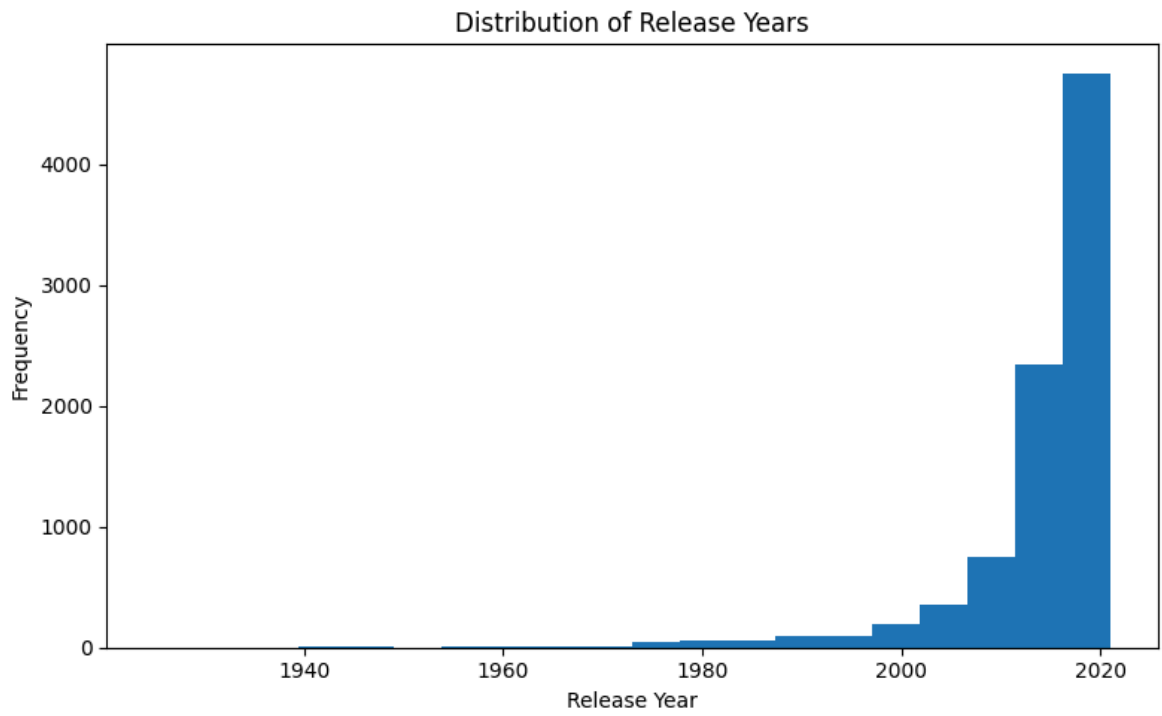
memory usage: 825.8+ KB

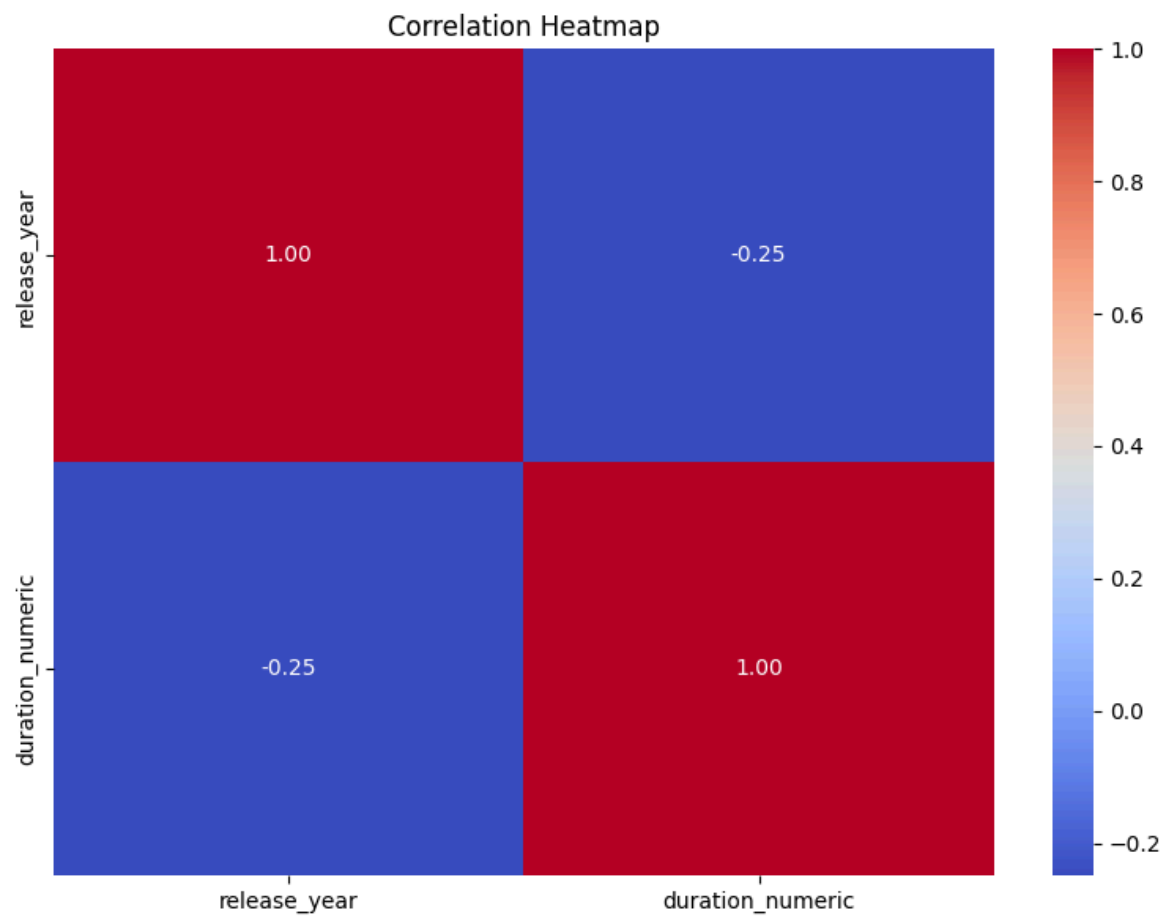
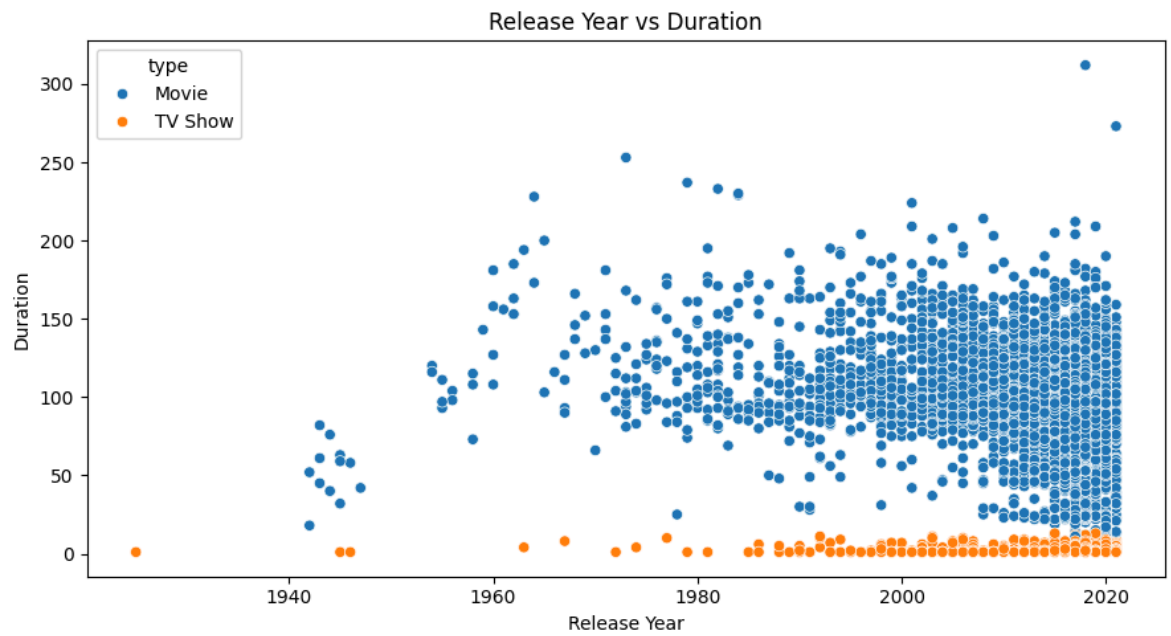
Missing Values:

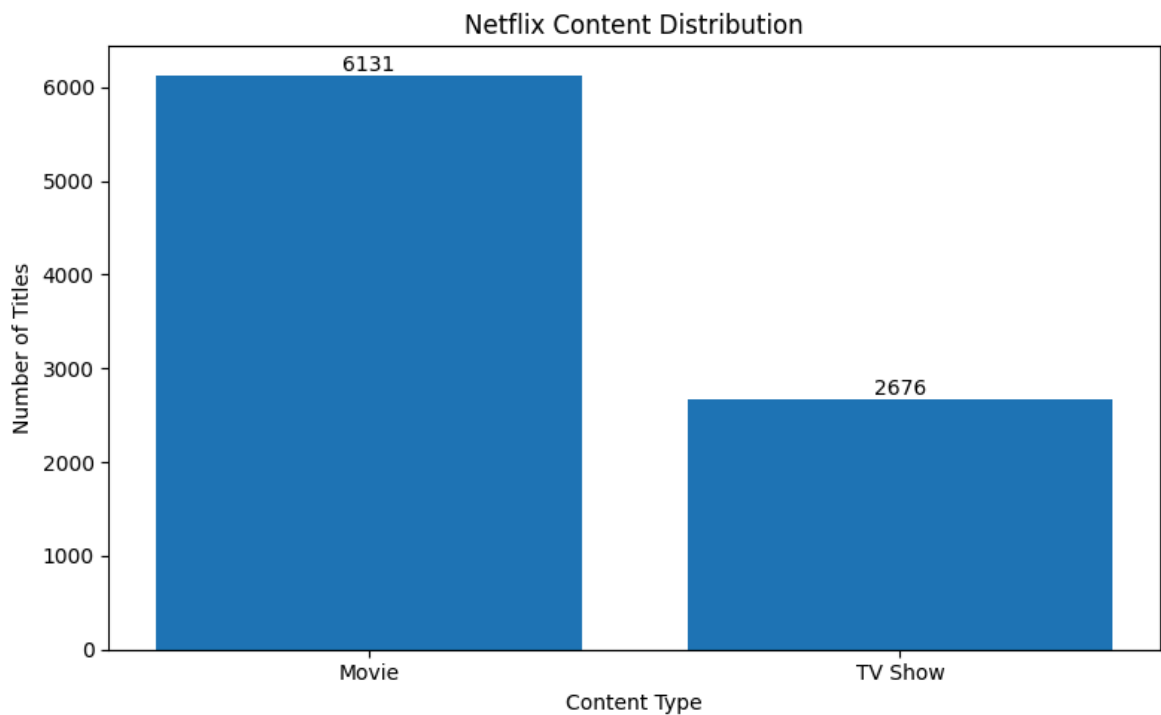
show_id	0
type	0
title	0
director	2634
cast	825
country	831
date_added	10
release_year	0
rating	4
duration	3
listed_in	0
description	0

dtype: int64









Visualization Summary

1. The bar chart compares Movies and TV Shows.
2. The line chart shows content distribution across release years.
3. The histogram shows the distribution of release years.
4. The box plot compares release-year distributions.
5. The scatter plot shows the relationship between release year and duration.
6. The heatmap shows correlations between numerical variables.
7. Customized charts improve readability and interpretation.

Experiment completed successfully!

Question

1. Why is data visualization considered an essential component of data analytics?

Data visualization converts data into graphical form, making it easier to:

- Identify trends and patterns.
- Find outliers.
- Understand relationships.
- Communicate insights clearly.
- Support decision-making.

2. Differentiate between Matplotlib and Seaborn. Which library is more suitable for statistical visualizations and why?

Matplotlib	Seaborn
General-purpose plotting library.	Statistical visualization library.
Provides detailed customization.	Provides attractive statistical plots.

Matplotlib	Seaborn
Requires more code for some plots.	Easier for statistical analysis.
Suitable for basic and advanced plots.	More suitable for statistical visualizations.

Seaborn is more suitable for statistical visualizations because it provides built-in statistical plots and works easily with Pandas DataFrames.

3. Which type of chart would you choose to compare sales across different regions? Justify your answer.

A **Bar Chart** is suitable for comparing sales across different regions.

It clearly shows the sales value of each region and makes it easy to identify the **highest and lowest sales**.

Example:

Region → Sales
East → 5000
West → 7000
North → 4500
South → 6000

4. What information can be obtained from a histogram and a box plot?

Histogram

Shows the **distribution of numerical data**.

It helps identify:

- Frequency.
- Distribution shape.
- Spread.
- Possible skewness.

Box Plot

Shows:

- Median.
- Quartiles.
- Data spread.
- Outliers.

5. Explain the purpose of a scatter plot. How does it help in identifying relationships between variables?

A **scatter plot** shows the relationship between two numerical variables.

It helps identify:

- Positive relationships.
- Negative relationships.
- Weak relationships.
- Clusters.
- Outliers.

For example, it can show the relationship between **advertising expenditure and sales**.

6. What is a correlation heatmap? How can it assist in feature selection for machine learning?

A **correlation heatmap** is a graphical representation of correlations between numerical variables.

It uses different shades to represent the strength and direction of relationships.

It helps in feature selection by identifying:

- Highly correlated features.
 - Redundant features.
 - Features strongly related to the target variable.
-

7. Why is chart customization (titles, labels, legends, colors) important in data visualization?

Chart customization improves **readability and understanding**.

It helps users:

- Understand the chart quickly.
 - Identify variables.
 - Compare categories.
 - Interpret results correctly.
 - Communicate insights effectively.
-

8. What factors should be considered while selecting an appropriate visualization for a dataset?

Important factors include:

- Type of data.
- Number of variables.
- Analytical objective.
- Relationship being studied.
- Audience.

- Amount of data.
- Need to show trends, comparisons, distributions, or relationships.

Examples:

Purpose	Chart
Comparison	Bar Chart
Trend	Line Chart
Distribution	Histogram
Relationship	Scatter Plot
Correlation	Heatmap

9. How can misleading visualizations affect business decision-making? Give a real-world example.

Misleading visualizations can create **incorrect interpretations** and lead to poor business decisions.

For example, if a sales graph starts its Y-axis at a high value instead of zero, a small increase in sales may appear very large.

Therefore, incorrect scales, missing information, or inappropriate charts can mislead decision-makers.

10. Explain how data visualization supports storytelling and business intelligence in organizations.

Data visualization helps organizations convert complex data into **simple and meaningful insights**.

It supports:

- Business reporting.
- Trend analysis.
- Performance monitoring.
- Decision-making.
- Identifying opportunities and problems.

Dashboards and charts help managers quickly understand business performance and make **data-driven decisions**.